

76. Global Considerations in Kidney Disease Near and Middle East · assets 1-8 / 8



Fig. 76.1 Map of Middle Eastern countries. Countries illustrated in light green are also known as “El Maghreb”; those in dark green, as “El Meshreq.” (Modified from a public domain image available at http://en.wikipedia.org/wiki/Middle_East) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Fig_76_1

Table 764		Availability, Funding, Negative, and Advisory for Kidney Care – Neighborhood Workforce Capacity in MENA Eastern Countries										
		Availability of BMT		Availability of DOR		Funding for Renal/HD		Negative Distribution of Registry		Advisory Group		Neighborhood Workforce Capacity (P90)
Country	Year	IG	OC	Non-Transplant (IG)	Transplant (OC)	Non-Transplant (IG)	Transplant (OC)	IG	OC	IG	OC	Neighborhood Workforce Capacity (P90)
Algeria	2014											12
Algeria	2015											13
Algeria	2016											14
Algeria	2017											15
Algeria	2018											16
Algeria	2019											17
Algeria	2020											18
Algeria	2021											19
Algeria	2022											20
Algeria	2023											21
Algeria	2024											22
Algeria	2025											23
Algeria	2026											24
Algeria	2027											25
Algeria	2028											26
Algeria	2029											27
Algeria	2030											28
Algeria	2031											29
Algeria	2032											30
Algeria	2033											31
Algeria	2034											32
Algeria	2035											33
Algeria	2036											34
Algeria	2037											35
Algeria	2038											36
Algeria	2039											37
Algeria	2040											38
Algeria	2041											39
Algeria	2042											40
Algeria	2043											41
Algeria	2044											42
Algeria	2045											43
Algeria	2046											44
Algeria	2047											45
Algeria	2048											46
Algeria	2049											47
Algeria	2050											48
Algeria	2051											49
Algeria	2052											50
Algeria	2053											51
Algeria	2054											52
Algeria	2055											53
Algeria	2056											54
Algeria	2057											55
Algeria	2058											56
Algeria	2059											

Table 76 4 part1

[illegible]

Table_76_6_part1

Country	Code	Year	Prevalence		Age-Standardized Prevalence	Percentage Change
			Standardized	Unstandardized		
			1990	2017	1990	2017
Bahrain	BD	2017	12.2 (12.2)	17.7 (17.7)	12.2 (12.2)	48.4%
Brazil	BR	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Canada	CA	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
China	CN	2017	0.5 (0.5)	0.5 (0.5)	0.5 (0.5)	0%
France	FR	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Germany	DE	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
India	IN	2017	0.5 (0.5)	0.5 (0.5)	0.5 (0.5)	0%
Iran	IR	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Israel	IL	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Japan	JP	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Lebanon	LB	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Malaysia	MY	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Mexico	MX	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Norway	NO	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Qatar	QA	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Russia	RU	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Saudi Arabia	SA	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
South Korea	KR	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Taiwan	TW	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
Thailand	TH	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
United Kingdom	GB	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
United States	US	2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%
World		2017	1.0 (1.0)	1.0 (1.0)	1.0 (1.0)	0%

Table_76_2_part1

Country		Availability of KRT		Availability of CKM	Funding for Medications/KRT	Availability and Distribution of Registry			Advocacy Group	Nephrology Workforce (PMP)				
		HD	PD	Kidney transplantation	Shared Decision	Choice Restricted (limited)	CKD	Dialysis	Kidney transplantation	CKD	AKI	KRT	Nephrologist	Nephrologist trainees
United Arab Emirates	2017												8.7	0.9
	2019												6.7	0.72
	2023												NA	0.5
Yemen	2017												0.22	1.87

Green and red boxes indicate that survey participants answered yes and no, respectively, to GKHA Questionnaire^{36,156}; gray, not available. In “Funding for Medications/KRT”; dark green, publicly funded by government but with some fees at the point of delivery; green, publicly funded by government and free at the point of delivery; light green, mix of public and private funding systems; yellow, multiple systems—programs provided by government, nongovernmental organizations, and communities.

AKI, Acute kidney injury; CKD, chronic kidney disease; CKM, Conservative kidney management; HD, hemodialysis; KRT, kidney replacement therapy; PD, peritoneal dialysis; PMP, per million population.

Table 76 4 part2

Table 1b. Group-specific Correlations in Several Patient Populations with Familial Medionemian Pain (Continued)						
	Present Anxiety Disorder	Phenotype	Anxiety Disorder	Any Anxiety Disorder	Associated Rashes and Disorders	Reference
Jonathan et al. ¹⁰	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0	Cutaneous disorders	Major et al. ¹⁰
Preissner et al. ¹¹	ANXIETY DISORDER (YFSS)	NER	NER	NER		Ayoub et al. ¹¹
Adams et al. ¹²	ANXIETY DISORDER (YFSS)	NER	NER	NER		Adams et al. ¹²
Adams et al. ¹³	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Edmond et al. ¹³
Wagner (2004) ¹⁴	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Sexton et al. ¹⁴
Eggen et al. ¹⁵	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Geisler et al. ¹⁵
Eggen et al. ¹⁶	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Geisler et al. ¹⁶
Sprer et al. ¹⁷	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Martin et al. ¹⁷
Lofgren et al. ¹⁸	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Sjöström et al. ¹⁸
Algarin, Hershman and Tinsley ¹⁹	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Balshin et al. ¹⁹
Algarin et al. ²⁰	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Alkirk et al. ²⁰
Turkmen et al. ²¹	ANXIETY DISORDER (YFSS)	Acute pain, low threshold	YFSS, 0.001 to 1.0 patients YFSS (n = 10)	0.001 to 1.0		Chakraborty et al. ²¹

Table_76_6_part2

Country	Treated Kidney Failure		Chronic Dialysis (HD+PD)		Chronic Hemodialysis		Chronic Peritoneal Dialysis		Kidney Transplantation				Date of Country's First KRT Modality ^a				
	Incidence (P/MP)	Prevalence (P/MP)	Prevalence (P/MP)	Centers (P/MP)	Prevalence (P/MP)	Centers (P/MP)	Prevalence (P/MP)	Centers (P/MP)	Incidence (P/MP)	Prevalence (P/MP)	Incidence of deceased (P/MP)	Incidence of living donor (P/MP)	Incidence of deceased (P/MP)	HD ^b R, A	PD ^b R, A	Transplant ^b R, A	
Algeria			251.4	8.22	240.3			11.1	0.31	2.07		52.7	0	2.07	—	1962	1962
Egypt	207.3	3.37	250.7											1971	1968	1968	
Iraq	56	60.4	61.0	5.83	51.7	0.03	0.1	0.32	14.9				0.149	—	1968	1962	
Israel	7.4	6.4	7.4	0.6	6.4	0.3	17.8	0.3	14.7	31.1	9.76	5	—	R	1965	1967	
Iraq	—	251	145	0.64	—	0.07	—	—	—	100	—	—	—	R	1967	2004	
Jordan	—	7.7	127.9	—	—	—	62.35	—	—	—	103.2	16.33	37.0	—	R	1964	1964
Jordan	—	—	446.3	7.91	428.9	0.5	1.74	1.55	10	—	—	—	0.1	9.1	R	1968	1990
Lebanon	351.1	99.9	—	—	—	—	—	—	—	99	432 ^c	—	—	—	—	1976	—
Lebanon	—	54.1	151	66.51	0.94	25.5	0.94	12.88	—	—	—	1.02	11.86	N	1980	1994	1972
Morocco	144	541	185.6	15	—	—	0.8	0.15	0.35	—	—	0.14	0.22	—	1972	2005	1969
Morocco	—	11.4	114.3	—	—	—	0.22	—	0.19	0.1	—	—	—	—	1976	1962	1970
Oman	120	670	356	6.64	250	2.66	10.9	0.53	2.22	279	0	2.22	N	1981	1996	1996	
Oman	—	2.67	3.03	—	—	—	—	—	—	—	—	—	—	—	—	1993	—
Qatar	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	1968	—
Saudi Arabia	164	645	366	2.39	239.1	1.2	64.5	0.4	16.55	279	5.86	10.69	0.27	N	1983	1996	1996
Saudi Arabia	183 ^c	653 ^c	653 ^c	5.66	609.8	0.71	35	0.28	31	188 ^c	2.58	27	N	1971	1984	1984	1984
Saudi Arabia	—	—	155.1	2.5	146.1	0.32	6	0.28	17.21	—	—	—	0.721	N	1974	1984	1976
Tunisia	242	119.8	15.89	72.88	17.0	0.76	36.3	0.1	11.06	—	1.09	0.88	—	—	1973	1985	1985
Turkey	150 ^c	190 ^c	10.86	84.4	—	—	—	0.32	1.11	—	244 ^c	—	2.95	—	—	1968	1968
United Arab Emirates	—	152	204.1	10.1	184.1	0.5	17.3	0.61	1.3	—	—	—	6.6	—	R	1977	1995
Yemen	—	—	106.4	—	99.1	—	9.3	—	—	—	52.7	—	—	—	1982	1984	1998

Source: U.S. Renal Data System 2003 Annual Data Report: International Comparisons. Available at <http://www.atsdrhhs.gov/2003/and-stage-and-label-11-national-comparisons/>.

Data source: References (1)–(10) and (12)–(16).

HD therapy for acute kidney injury or end-stage kidney disease; PD, peritoneal dialysis; P/MP, per capita million; R, national; R, regional – data not reported or unavailable.

From Delo et al., 2003. *GBA Global Health Atlas*. San, Brussels. Available at www.broton.org/global atlas.

Table_76_2_part2

Table 2.9. The Genetic Clones in Phenotyping Report in Mideast Eastern Europe						
Category	Gene	GWAS No.	Phenotype	Effects, Mutation	References	
Grain yield	Arabidopsis thaliana	20000	Grain yield, seed weight, seed number, seed volume	A191, A192, A193, A194, A195, A196, A197, A198, A199, A200, A201, A202, A203, A204, A205, A206, A207, A208, A209, A210, A211, A212, A213, A214, A215, A216, A217, A218, A219, A220, A221, A222, A223, A224, A225, A226, A227, A228, A229, A230, A231, A232, A233, A234, A235, A236, A237, A238, A239, A240, A241, A242, A243, A244, A245, A246, A247, A248, A249, A250, A251, A252, A253, A254, A255, A256, A257, A258, A259, A260, A261, A262, A263, A264, A265, A266, A267, A268, A269, A270, A271, A272, A273, A274, A275, A276, A277, A278, A279, A280, A281, A282, A283, A284, A285, A286, A287, A288, A289, A290, A291, A292, A293, A294, A295, A296, A297, A298, A299, A300, A301, A302, A303, A304, A305, A306, A307, A308, A309, A310, A311, A312, A313, A314, A315, A316, A317, A318, A319, A320, A321, A322, A323, A324, A325, A326, A327, A328, A329, A330, A331, A332, A333, A334, A335, A336, A337, A338, A339, A340, A341, A342, A343, A344, A345, A346, A347, A348, A349, A350, A351, A352, A353, A354, A355, A356, A357, A358, A359, A360, A361, A362, A363, A364, A365, A366, A367, A368, A369, A370, A371, A372, A373, A374, A375, A376, A377, A378, A379, A380, A381, A382, A383, A384, A385, A386, A387, A388, A389, A390, A391, A392, A393, A394, A395, A396, A397, A398, A399, A400, A401, A402, A403, A404, A405, A406, A407, A408, A409, A410, A411, A412, A413, A414, A415, A416, A417, A418, A419, A420, A421, A422, A423, A424, A425, A426, A427, A428, A429, A430, A431, A432, A433, A434, A435, A436, A437, A438, A439, A440, A441, A442, A443, A444, A445, A446, A447, A448, A449, A450, A451, A452, A453, A454, A455, A456, A457, A458, A459, A460, A461, A462, A463, A464, A465, A466, A467, A468, A469, A470, A471, A472, A473, A474, A475, A476, A477, A478, A479, A480, A481, A482, A483, A484, A485, A486, A487, A488, A489, A490, A491, A492, A493, A494, A495, A496, A497, A498, A499, A500, A501, A502, A503, A504, A505, A506, A507, A508, A509, A510, A511, A512, A513, A514, A515, A516, A517, A518, A519, A520, A521, A522, A523, A524, A525, A526, A527, A528, A529, A530, A531, A532, A533, A534, A535, A536, A537, A538, A539, A540, A541, A542, A543, A544, A545, A546, A547, A548, A549, A550, A551, A552, A553, A554, A555, A556, A557, A558, A559, A560, A561, A562, A563, A564, A565, A566, A567, A568, A569, A570, A571, A572, A573, A574, A575, A576, A577, A578, A579, A580, A581, A582, A583, A584, A585, A586, A587, A588, A589, A590, A591, A592, A593, A594, A595, A596, A597, A598, A599, A600, A601, A602, A603, A604, A605, A606, A607, A608, A609, A610, A611, A612, A613, A614, A615, A616, A617, A618, A619, A620, A621, A622, A623, A624, A625, A626, A627, A628, A629, A630, A631, A632, A633, A634, A635, A636, A637, A638, A639, A640, A641, A642, A643, A644, A645, A646, A647, A648, A649, A650, A651, A652, A653, A654, A655, A656, A657, A658, A659, A660, A661, A662, A663, A664, A665, A666, A667, A668, A669, A670, A671, A672, A673, A674, A675, A676, A677, A678, A679, A680, A681, A682, A683, A684, A685, A686, A687, A688, A689, A690, A691, A692, A693, A694, A695, A696, A697, A698, A699, A700, A701, A702, A703, A704, A705, A706, A707, A708, A709, A710, A711, A712, A713, A714, A715, A716, A717, A718, A719, A720, A721, A722, A723, A724, A725, A726, A727, A728, A729, A730, A731, A732, A733, A734, A735, A736, A737, A738, A739, A740, A741, A742, A743, A744, A745, A746, A747, A748, A749, A750, A751, A752, A753, A754, A755, A756, A757, A758, A759, A760, A761, A762, A763, A764, A765, A766, A767, A768, A769, A770, A771, A772, A773, A774, A775, A776, A777, A778, A779, A780, A781, A782, A783, A784, A785, A786, A787, A788, A789, A790, A791, A792, A793, A794, A795, A796, A797, A798, A799, A800, A801, A802, A803, A804, A805, A806, A807, A808, A809, A810, A811, A812, A813, A814, A815, A816, A817, A818, A819, A820, A821, A822, A823, A824, A825, A826, A827, A828, A829, A830, A831, A832, A833, A834, A835, A836, A837, A838, A839, A840, A841, A842, A843, A844, A845, A846, A847, A848, A849, A850, A851, A852, A853, A854, A855, A856, A857, A858, A859, A860, A861, A862, A863, A864, A865, A866, A867, A868, A869, A870, A871, A872, A873, A874, A875, A876, A877, A878, A879, A880, A881, A882, A883, A884, A885, A886, A887, A888, A889, A890, A891, A892, A893, A894, A895, A896, A897, A898, A899, A900, A901, A902, A903, A904, A905, A906, A907, A908, A909, A910, A911, A912, A913, A914, A915, A916, A917, A918, A919, A920, A921, A922, A923, A924, A925, A926, A927, A928, A929, A930, A931, A932, A933, A934, A935, A936, A937, A938, A939, A940, A941, A942, A943, A944, A945, A946, A947, A948, A949, A950, A951, A952, A953, A954, A955, A956, A957, A958, A959, A960, A961, A962, A963, A964, A965, A966, A967, A968, A96		

Table 76.5